

Question 1: Student–Department Table (STUDENT_INFO)

1. **Candidate Key:** {stud_id}.
2. **Dependencies:**
 - **Partial:** None.
 - **Transitive:** stud_id -> dept_id -> hod_id -> hod_name. **Violates 3NF.**
3. **Normalization Steps (1NF -> 3NF):**
 - Decomposition 1: Remove hod_id -> hod_name.
 - Decomposition 2: Remove dept_id -> dept_name, hod_id.
4. **Final 3NF Tables:**
 - **HOD:** (hod_id [PK], hod_name)
 - **DEPARTMENT:** (dept_id [PK], dept_name, hod_id [FK])
 - **STUDENT:** (stud_id [PK], stud_name, dept_id [FK])

Question 2: Employee–Project Table (EMP_PROJECT)

1. **Primary Key (PK):** {emp_id, proj_id}.
2. **Violations of 3NF:**
 - **2NF Violation (Partial):** emp_id -> emp_name and proj_id -> proj_name, client_id.
 - **3NF Violation (Transitive):** client_id -> client_name (in the resulting project table).
3. **Normalization Steps (1NF -> 2NF -> 3NF):**
 - 1NF -> 2NF: Separate employee and project details from the link table.
 - 2NF -> 3NF: Decompose PROJECT table using client_id -> client_name.
4. **Final Tables:**
 - **CLIENT:** (client_id [PK], client_name)
 - **PROJECT:** (proj_id [PK], proj_name, client_id [FK])
 - **EMPLOYEE:** (emp_id [PK], emp_name)
 - **ASSIGNMENT:** (emp_id [PK, FK], proj_id [PK, FK])

Question 3: Library Books Table (LIBRARY)

1. **All FDs:** book_id -> book_title, author_id, publisher_id; author_id -> author_name; publisher_id -> publisher_name.
2. **Transitive Dependencies (PK: book_id):**
 - book_id -> author_id -> author_name.

- book_id -> publisher_id -> publisher_name. **Violates 3NF.**
- 3. **Normalization to 3NF:** Separate author and publisher details.
- 4. **Final 3NF Tables:**
 - **AUTHOR:** (author_id [PK], author_name)
 - **PUBLISHER:** (publisher_id [PK], publisher_name)
 - **BOOK:** (book_id [PK], book_title, author_id [FK], publisher_id [FK])

Question 4: Sales Table (SALES)

1. **Transitive Dependencies (PK: sale_id):**
 - sale_id -> product_id -> category -> category_discount. **Violates 3NF.**
2. **Normalization up to 3NF:** Separate the dependencies into tables based on product_id and category.
3. **Final Schema:**
 - **CATEGORY_DISCOUNT:** (category [PK], category_discount)
 - **PRODUCT:** (product_id [PK], product_name, category [FK])
 - **SALE:** (sale_id [PK], product_id [FK])

Question 5: Student Course Enrollment (ENROLL)

1. **Partial Dependencies (PK: {roll_no, course_id}):**
 - roll_no -> stud_name.
 - course_id -> course_name, credits. **Violates 2NF.**
2. **Normalization to 3NF:** Separate non-key attributes that depend on only part of the composite key.
3. **Clean Table Set:**
 - **COURSE:** (course_id [PK], course_name, credits)
 - **STUDENT:** (roll_no [PK], stud_name)
 - **ENROLLMENT:** (roll_no [PK, FK], course_id [PK, FK])

Question 6: Course–Faculty Table (COURSE_FACULTY)

1. **Candidate Key:** {course_code}.
2. **Check 3NF:** The table is in 3NF.
3. **Identify BCNF Violation:** Determinants faculty_id and room_no are **NOT** superkeys. FD: room_no -> faculty_id. **Violates BCNF.**

4. **Decompose into BCNF:** Decompose the violating FDs.
 - **FACULTY:** (faculty_id [PK], faculty_name)
 - **ROOM_ASSIGNMENT:** (room_no [PK], faculty_id [FK])
 - **COURSE_SCHEDULE:** (course_code [PK], course_name, room_no [FK])

Question 7: Orders Table (ORDERS)

1. **Check 2NF & 3NF:** In 2NF, but **NOT** in 3NF (Transitive dependency: city -> pincode).
2. **Identify BCNF Violations:** After 3NF, the CUSTOMER table violates BCNF: city -> pincode, where city is not the PK.
3. **Convert to BCNF:**
 - **CITY_PINCODE:** (city [PK], pincode)
 - **CUSTOMER:** (cust_id [PK], cust_name, city [FK])
 - **ORDER_HEADER:** (order_id [PK], cust_id [FK])

Question 8: Hostel Room Allocation (HOSTEL)

1. **Candidate Key:** {student_id}.
2. **Check 3NF:** **NOT** in 3NF (Transitive dependency: student_id -> room_no -> warden_id -> warden_name).
3. **Identify BCNF Violations:** After 3NF, the ROOM_WARDEN table violates BCNF: warden_id -> warden_name, where warden_id is not the PK.
4. **Normalize to BCNF:**
 - **WARDEN:** (warden_id [PK], warden_name)
 - **ROOM_ALLOCATION:** (room_no [PK], warden_id [FK])
 - **STUDENT_HOSTEL:** (student_id [PK], student_name, room_no [FK])

Question 9: Hospital Table (HOSPITAL)

1. **Candidate Key:** {doc_id, patient_id}.
2. **Normalize up to BCNF:** Decomposition required for 2NF (Partial) and 3NF (Transitive dependency on room_no).
3. **Final BCNF Tables:**
 - **DOCTOR:** (doc_id [PK], doc_name)
 - **ROOM_TYPE:** (room_no [PK], room_type)

- **PATIENT:** (patient_id [PK], patient_name, disease, room_no [FK])
- **TREATMENT:** (doc_id [PK, FK], patient_id [PK, FK])